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Real-Time Detection of Stage Isolation Failure Using Pressure Analysis in Multistage Hydraulic Fracturing

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Abstract

Achieving effective stage isolation during multistage hydraulic fracturing is essential to ensuring optimal stimulation coverage, efficient proppant placement, and production performance. Traditional diagnostics for stage isolation failure such as plug slip, plug leak, failure to seat the ball, or mechanical degradation have relied heavily on subjective manual interpretations or post-treatment data, offering limited value for real-time decision-making. This study introduces a novel, real-time, automated detection framework that integrates multiple pressure analysis techniques to identify isolation or plug integrity failures with over 95% accuracy during treatment operations.

The system evaluates and integrated six key criteria: (a) high frequency water hammer analysis with tubewave velocity calculations, (b) ISIP/falloff signature alignment, (c) sharp intra-stage pressure drops, (d) ball-seat detection and validation, (e) hammer frequency anomalies, and (f) pressure behavior at design rate. These pressure responses are segmented across operational cycles using event detection logic and analyzed independently through algorithmic logic. The individual outputs are combined through a multi-criteria decision analysis using a weighted rubric informed by the Analytical Hierarchy Process (AHP), which adjusts based on completion type (e.g., cased hole plug-and-perf vs. openhole MSF) and treatment mode (proppant vs. acid).

This diagnostic engine not only identifies isolation loss but also distinguishes between plug damage and mechanical seating issues, with results validated against SME interpretation. The workflow provides operational teams with actionable insights, reducing non-productive time and enabling preemptive corrections. This is the first known integration of real-time multi-model pressure analytics with digital logic automation for isolation diagnostics in field treatments, offering an approach with near-100% accuracy for plug performance monitoring and frac design optimization, with a potential to save 10s to 100s operational hours.

Introduction and Background

In multistage horizontal hydraulic fracturing, mechanical isolation between stages is typically achieved using frac plugs designed to prevent fluid communication with previously stimulated intervals. When these plugs fail due to improper seating, mechanical damage, or erosion, interstage communication can occur, resulting in loss of isolation. This failure leads to the diversion of fracturing fluid into an already-stimulated zone, thereby reducing treatment efficiency and compromising overall production performance. In openhole MSF completions, this can result from hydraulic communication within the reservoir, ball failure or packer failures. In cased hole plug-n-perf wells, it results from hydraulic communication within the reservoir, ball failure or plug failure. Fig. 1 illustrates both successful isolation and a scenario of communication, where fluid leaks back into the previously treated stage. The success of multistage hydraulic fracturing is fundamentally dependent on effective zonal isolation between stages. Whether utilizing plug-and-perf or openhole multistage fracturing (MSF), ensuring that each stage is independently stimulated prevents unwanted fluid leakoff, improves cluster efficiency, and ensures accurate proppant placement. As lateral lengths and stage densities continue to increase, reliable detection of stage isolation failure becomes critical, especially since such failures can lead to reduced production, casing damage, and significant non-productive time (Peña et al. 2025).

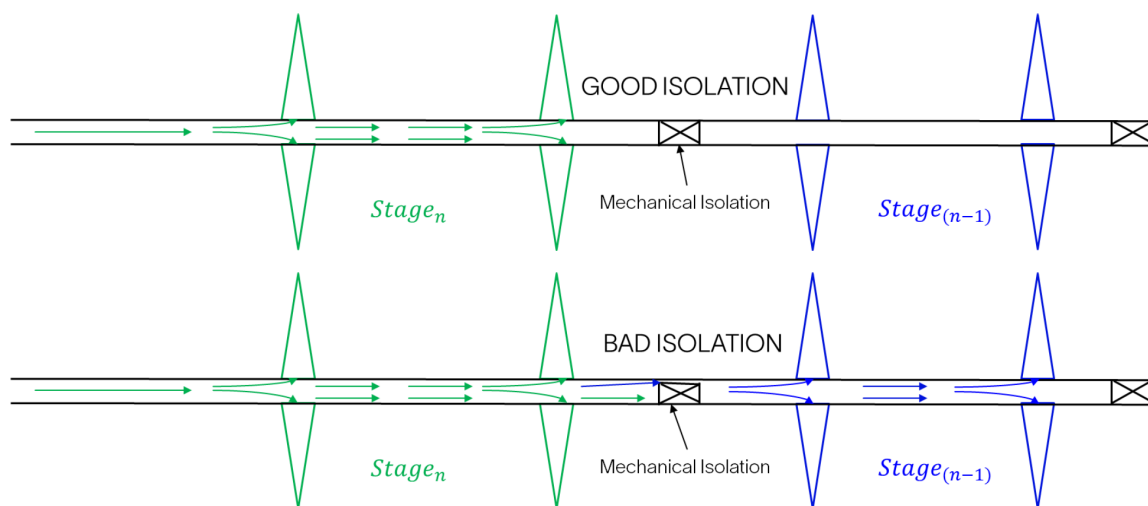


Figure 1—Stage isolation in context of mechanical plug isolation (top) showing good isolation (bottom) showing bad isolation or lack of isolation. Stage n is indicated by the currently treated stage and stage (n-1) indicates the past stimulated stage

Historically, verification of effective isolation relied on indirect indicators such as fracture geometry interpretation, post-job diagnostics (e.g., tracers or production logs), or time-consuming manual pressure trend analyses. Advanced diagnostics like distributed acoustic sensing (DAS), real-time video perforation imaging, and water hammer analysis have provided new insights, yet their application remains limited by cost, infrastructure, or processing delays. Recent studies have highlighted both the need and opportunity to leverage surface pressure signatures and pump-down diagnostics to infer stage-to-stage isolation integrity with high reliability. Cramer et al. (2020) introduced pump-down diagnostic techniques capable of evaluating plug integrity and cement sheath effectiveness through pressure response analysis during wireline operations. Their study emphasized that plug-seating verification and cross-stage communication trends could be inferred economically without the need for downhole hardware. Similarly, Savitski et al. (2024) presented an integrated fiber optic pilot in the Vaca Muerta, demonstrating that discrepancies between Isolation Integrity Tests and DAS/DTS interpretations can reveal hidden isolation issues, even in cases where plug-seating signatures appear positive. They found that isolation challenges often manifest later in treatment and are sometimes masked by surface pressure alone, underscoring the importance of combining

multiple diagnostic perspective. Verrastro et al. (2025) expanded this concept by evaluating over 700 stages in Vaca Muerta, confirming that pre-treatment water hammer analysis, step-down rate tests, and ISIP trend comparisons could predict isolation failure with up to 81% accuracy. Their approach demonstrated the potential of surface-based pressure diagnostics to perform on par with more invasive downhole tools, when implemented systematically. Baki et al. (2023) also conducted a large-scale evaluation of over 6,000 frac plugs to assess their impact on mill-out efficiency and well performance in unconventional reservoirs. The study highlights the operational significance of choosing the appropriate plug type (composite, dissolvable, or semi-dissolvable) based on downhole conditions, milling requirements, and debris handling. Composite plugs emerged as highly efficient in terms of milling time and operational stability, while dissolvable plugs offered initial advantages that diminished after prolonged downhole exposure. The paper also emphasizes the critical importance of systematic plug qualification workflows, including trial testing and acid dissolution assessments, to avoid complications such as interstage communication or plug failure during operations. These findings underscore the operational challenges that can indirectly lead to stage isolation issues, especially when plugs fail to dissolve or mill as intended. The broader push for 100% stage isolation has also been reinforced by Peña et al. (2025), who examined the performance of various frac plug technologies. Their work showed that even when plugs are tagged during mill-out, isolation cannot be assumed. Real-time tools capable of distinguishing between mechanical integrity and functional isolation are thus essential to validate each stage during treatment

In this context, the present study introduces a real-time digital workflow that applies various independent pressure analysis techniques, ranging from ISIP trend analysis and water hammer diagnostics to pressure drop slopes and friction gradient comparison, to detect isolation loss and plug performance failure. The methodology further incorporates an MCDA engine using the Analytical Hierarchy Process (AHP), which assigns adaptive weights to each diagnostic metric based on the well completion type and treatment method. This integrated logic-based system bridges the gap between post-job interpretation and real-time action, offering operators a scalable and automated solution for stimulation efficiency assurance. For instance, the integrated analysis would take up to a few hours after each stage and pumping event to conduct the analysis alone. Adding to that, data transfer and peer review with the decision makers adds another few hours. If pumping is occurring during the night, sometimes decisions cannot be made until daylight. Here, the real-time answer product with agreed metrics and decision making workflows could yield the result immediately and yield a choice for the engineers to run an additional plug or either to skip the stage, avoiding significant non-productive time (NPT), while securing the technically-weighted decision with completely engineered analysis.

Methodology

In this study, we demonstrate the deployment of an automated digital solution that uses algorithmic logic encoding domain science and principles to automatically detect the mechanical or chemical plug integrity and stage/interval isolation. After the automated event detection, a scientific multi-criteria decision analysis (MCDA) workflow has been layered to provide the answer.

Solution Approach

A structured diagnostic framework has been developed to assess the integrity of stage isolation during multistage hydraulic fracturing. The foundation of the workflow is visualized in Fig. 2, which delineates the key pressure-based diagnostic events associated with the ongoing treatment stage (stage n , shown in green) and the previously stimulated stage ($n-1$, shown in blue). To operationalize this, the approach utilizes changes in pressure baselines, slope analysis, and pressure-wave signatures across these markers. This perspective enables cross-stage comparisons that are essential for detecting communication due to plug failure or loss of isolation. Each stage contributes distinct data features such as instantaneous shut-in

pressure (ISIP), stable pad pressures at design rate, and decline trends which are interpreted in a temporal and comparative context to assess stage-to-stage hydraulic communication (or lack thereof).

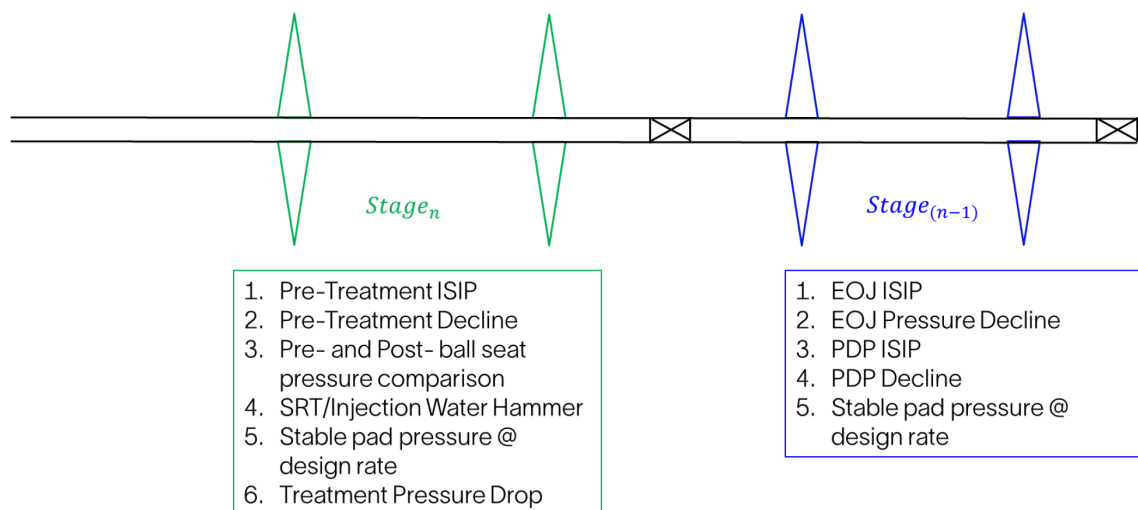


Figure 2—Schematic of the ongoing stage-n (green) and previous stage (n-1) (blue) and pressure diagnostic components utilized for comparative and integrated stage/interval isolation analysis. EOJ: end of job; ISIP: Instantaneous shut-in pressure PDP: Pump down performing.

For the ongoing stage n , the workflow incorporates six signals and for the preceding stage $n-1$, five data points are extracted post-job (depicted in [figure 2](#)). These events capture the full evolution of pressure behavior before and during stimulation, providing high-fidelity diagnostic capability. Utilizing multiple indicators together enhances confidence in the analysis and decision making.

The other important question is the possibility of a plug failure at different instances in the operational lifecycle of a single stage. Hence, [Fig. 3](#) further exemplifies the sub-cycle analysis workflow across consecutive stages, including main fracturing (for stages $n-1$ and n), PDP, and injection diagnostics (SRT/Mini Fall-Off). The evolution of pressure trends in time-series provide critical insight into when, and how, isolation loss manifests.

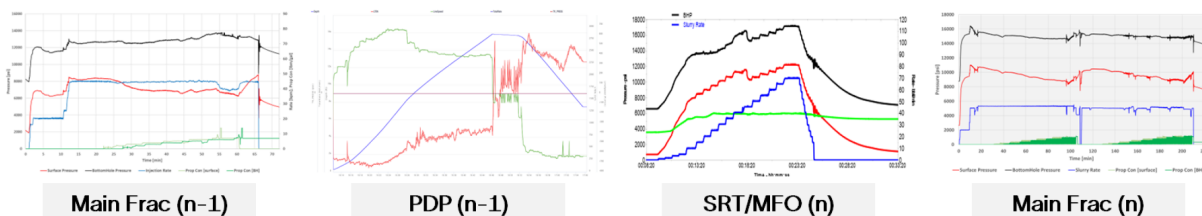


Figure 3—Schematic showing multiple instances of stage isolation loss at stages n and $(n-1)$.

Complementing the figure, [Table 1](#) summarizes the three main instances of isolation failure, each mapped to the diagnostic stage where detection is most probable.

- In Instance 1, where the plug fails to set entirely, the failure is typically diagnosed during the PDP cycle of the preceding stage ($n-1$) by identifying the absence of expected pressure test performance or pressure build-up.
- Instance 2 involves cases where the plug is set but fails hydraulically before the main stimulation of stage n . These are most reliably diagnosed using pre-treatment injection and water hammer analysis, which may show abnormal pressure behavior and lack of resistance expected from a virgin rock which has not been broken down.

- Instance 3 represents plug integrity failure during the active fracturing of stage n , often identifiable only during main treatment pressure monitoring or post-shut-down decline diagnostics.

Table 1—Loss of isolation depiction with pumping cycles

#	Isolation loss instance	Identified at which step?
1	Plug did not set and isolation not achieved	Pressure analysis during PDP
2	Plug set without issues but fails before the main treatment starts	Pre-treatment diagnostic tests such as SRT, Injection, etc.
3	Plug fails during the main treatment	During main treatment and after pumps shut down

The classification in [Table 1](#) informs automated logic for real-time alerts and provides operators with actionable insights based on the failure instance without requiring additional instrumentation. The early detection of isolation issues enables timely remediation, contingency matrix activation or engineered adjustments to the treatment.

Fracture Pressure Criteria

To ensure robust and high-resolution stage isolation diagnostics, a suite of pressure-based criteria has been developed and encoded in the digital solution, as summarized in [Table 2](#). These metrics serve as the core indicators and evaluation metrics of hydraulic integrity between successive stages during multistage fracturing operations. Each diagnostic is associated with a quantifiable pressure signature, a temporal or amplitude-based threshold, and a specific treatment sub-cycle where the signal is most reliably observed. The workflow allows for modular customization of threshold values depicted by ‘x’, ‘y’, etc. in the table (user experience component aided by a reinforcement learning based ML feedback loop), enabling adaptation to different reservoir types, wellbore designs, and geographical basins.

Table 2—Detailed fracture pressure criteria selection and analysis.

	Metric	Pumping Cycle for Identification	Description	Quantifiable Measure
1	ISIP	Post shut-in after SRT and main treatment	ISIPs overlap when fluid is entering the same zone	ΔP_{ISIP}
2	Decline	Post shut-in after SRT and main treatment	First 5-min decline rate will overlap if leakoff is dominated by the same interval	$\frac{dP}{dt}$
3	Pressure Drop	During main treatment	A drastic pressure drop typically implies near-wellbore restriction removal or plug failure	ΔP_{treat}
4	Drop Derivative	During the observation of the sharp drop within main treatment	A sharper drop usually implies mechanical failure	$\frac{dP}{dt} \bigg _{t=t_0}^{t=t_0+y}$
5	Ball Seat Pressure Comparison	Ball landing – injection or SRT	Pre- and post-ball seating/landing pressure should show a sizeable differential if isolation is successful	ΔP_{seat}
6	Pressure at design rate	SRT or main treatment pad step	NWB pressure loss component is significant during injection and pad steps. So, If the pad pressure is low, it could be dominated by eroded perfs from previous stage	$\Delta \left(\frac{dP}{dQ} \right)$
7	Water Hammer Frequency	Post shut-in after SRT and main treatment	The frequency of the water hammer (no. of oscillations) in pre-frac diagnostic dictates the near wellbore connectivity	$f < 4$ oscillations – good isolation; $f = 4-7$ – transition zone; $f > 7$ oscillations - lack of isolation

1. ISIP Overlap: A critical signature for stage communication is the similarity between ISIPs of consecutive stages. A large overlap in ISIP values between stages n and $n-1$ suggests that the injection

fluid is encountering similar rock behavior and restrictions, potentially due to communicating fractures. Computed as in [equation 1](#), where values of $|\Delta P_{ISIP}|$ less than a predefined threshold, ϵ_1 (typically 100-200 psi) flag possible loss of isolation.

$$\Delta P_{ISIP} = ISIP_n - ISIP_{n-1} \quad (1)$$

2. Initial Decline Behavior: The early-time decline following pump shut-down provides valuable insights into fracture conductivity and connectivity. If the first 5-minute slope of stage n mimics that of $n-1$, or for following pumping cycles within the stage, it suggests fluid loss to the same zone/interval. The derivatives are computed as in expression 2, where quantitative comparison between cycles is used as an indicator with custom slope derivative thresholds (ϵ_2). Note that for comparing significantly different fluid volumes, we use specialized G-function time instead of time for more representative comparisons.

$$\frac{dP}{dt} \Big|_{t=0-5min} \text{ OR } \frac{dP}{dG} \Big|_{t=0-5min} \quad (2)$$

3. Treatment Pressure Drop: Abrupt drops in treatment pressure may signal a mechanical failure, such as plug slip or mechanical deformation. The metric is shown in [equation 3](#), where x is user-defined

$$\Delta P_{treat} = P_{initial} - P_{stable} \text{ } | x \text{ seconds} \quad (3)$$

4. Treatment Pressure Drop Derivative: An extension of the pressure drop analysis, the custom defined derivative threshold (ϵ_3) in [equation 4](#) examines the rate of decline over a defined interval y , where a rapid drop derivative indicates dynamic leak-off or mechanical failure. can be custom defined (typically > 1500 psi)

$$\epsilon_3 = \frac{dP}{dt} \Big|_{t=t_0}^{t_0+y} \quad (4)$$

5. Ball Seat Pressure Comparison: Critical measure for ball drop completions, the pressure signature across ball seating and landing serves as a marker for mechanical integrity. A successful seat should generate a pressure spike due to instantaneous change in flow area. Failure to observe this spike implies plug malfunction or bypass. The metric can be quantified as per [equation 5](#) where values of ΔP_{seat} less than a predefined threshold, ϵ_4 (typically 100-200 psi) flag possible loss of isolation.

$$\Delta P_{seat} = P_{post-seat} - P_{pre-seat} \quad (5)$$

6. Pad Pressure at Design Rate: By comparing the pressure required to maintain same injection rate across stages, insights into flow resistance and eroded perforations from previous stage can be obtained. This criterion is better suited for horizontal wells where lateral homogeneity can be assumed.

The metric can be quantified as per [equation 6](#) where anomalously low $\frac{dP}{dQ}$ in stage n indicates lower flow resistance implying plug leakage or flow into already eroded perforation of stage $n-1$.

$$\Delta \left(\frac{dP}{dQ} \right) = \frac{dP}{dQ}_n - \frac{dP}{dQ}_{n-1} \quad (6)$$

7. Water Hammer Oscillation Frequency: The frequency of water hammer oscillations during shut-in reflects near-wellbore hydraulic impedance as per [equation 7](#), where f is the water hammer frequency, c is the speed of sound in fluid, D is the effective depth to reflection point, k is the bulk modulus, and ρ is the density of fluid. For ease here, we estimate the proxy of isolation with the number of discernible cycles indicating the degree of reflection. The output is classified into a low, medium, high bin to allocate interpretations for good isolation, transition zone, and loss of isolation.

$$f = \frac{c}{4D}, \text{ where } c = \sqrt{\frac{k}{\rho}} \quad (7)$$

Automated Event Detection

Automated event detection becomes essential to extract relevant information from historical operations containing real-time timeseries data. Manual data interpretation is not feasible due to the inefficiencies and inconsistencies coming from the human factor introducing errors and bias to the data. The developed event detection service leverages built deterministic algorithms incorporating domain-specific logics to identify critical operational events. These algorithms provided a reliable methodology to detect target events, assess the events, extract information and deliver the outcome result providing consistent results removing human prone errors and biases.

Prior to running the algorithms, preprocessing is required to ensure the raw data that includes timeseries data are free from null values, noise, and format inconsistencies that could compromise the integrity of the algorithms computations. Ensuring homogeneity in the data boosts the reliability of the event detection service by minimizing the algorithms failure risk. Thus, enhancing the data extraction effectiveness.

Each of the metrics mentioned in the fracture pressure criteria is automatically detected through algorithmic logics based on the quantifiable measures. Each of the process returns a binary answer. The different metrics are evaluated at different treatment periods. Fig. 4 illustrates the workflow for evaluating stage isolation using information from different treatment periods, using the aforementioned metrics.

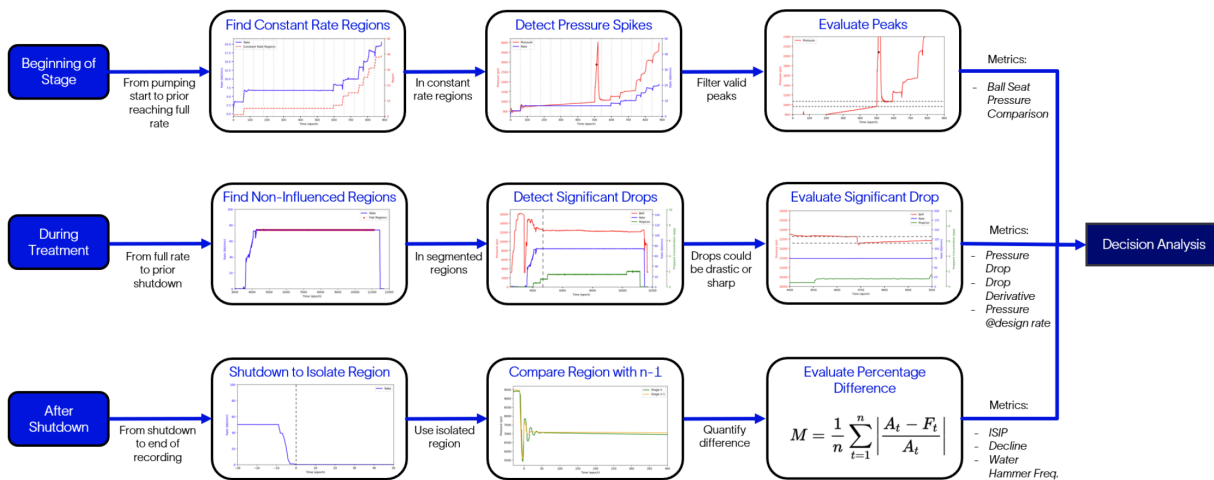


Figure 4—Automated Event Detection algorithm workflow showing the periods of the treatment whether the algorithm runs searching for any significant event based on the fracture pressure criteria. After evaluating the different periods of the stage, the results are assessed to conclude whether isolation was achieved or lost.

Beginning of the stage. At the beginning of the stage, prior to reaching design rate, the ball seating can be detected. During operations, isolation balls are pumped downhole to set against the ball seat creating a seal that isolates the previous stimulated cluster from the newly target cluster section to stimulate. However, ball seating is susceptible to several failure modes that can compromise stage isolation. Seat erosion, debris interference, improper pressure management can all prevent the ball from forming an effective seal. Such failures can result in cross-stage communication reducing stimulation efficiency.

For automated detection, evaluating both pre- and post-landing to identify any pressure differential provides information regarding the isolation effectiveness. When a ball is pumped and no signal is generated over more than a casing volume, it is a clear indication that the ball did not properly isolate the previous section. Another case is when the ball seats and pressure differential is generated, but the pressure after ball seating is the same as the pressure before ball arrives at the seat location. Ideally, when the ball seats a pressure differential should be generated and the pressure post-landing should be greater than the pressure pre-landing due to the hydraulic isolation from an already stimulated stage.

The automated event detection algorithm for ball seat detection isolates the region from start of pumping until before reaching design rate. This is our interval of interest as the ball is not pumped at design rate because the chances of ball failure and completion jewelry failure are high, at high rate. The first step of the algorithm logic is finding constant rate regions. This is done by calculating the derivatives of rate over time and clustering them in groups based on the derivative value being below a defined threshold. Fig. 5 shows the cluster regions given a rate during a ball seat landing. Ball seat detection can only occur in larger flat regions given a margin.

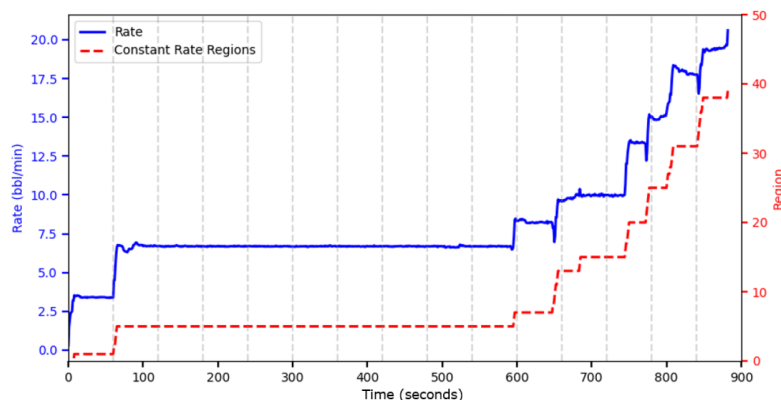


Figure 5—Constant rate regions given a slurry rate during a ball seat landing.

In constant rate regions, now we focus on the surface pressure behavior. During the constant rate, pressure should be constant with a slight increasing behavior. However, when a ball seats properly, a significant pressure spike will occur. This particular pressure signal is captured by looking at large pressure differentials. The developed algorithm logic will search for significant peaks filtering non-valid peaks using the statistical variation in the constant region detecting the most significant peak as observed in Fig. 6. After finding the most significant peak in the constant rate region, the peak result can be evaluated to assess whether or not isolation was achieved successfully sitting the ball prior to stimulating the stage. The algorithm searches to the left and right of the detected peak. Then it looks for a sharp reduction in the derivatives, meaning low variation within the desired timeframe where the point should be found. This is done prior, and post ball landing, capturing the pressure differential caused by the ball seating as observed in Fig. 6. Evaluating the pressure differential dependent on the information from previous and next stage can aid in calculating a quantitative metric that helps concluding whether isolation was achieved.

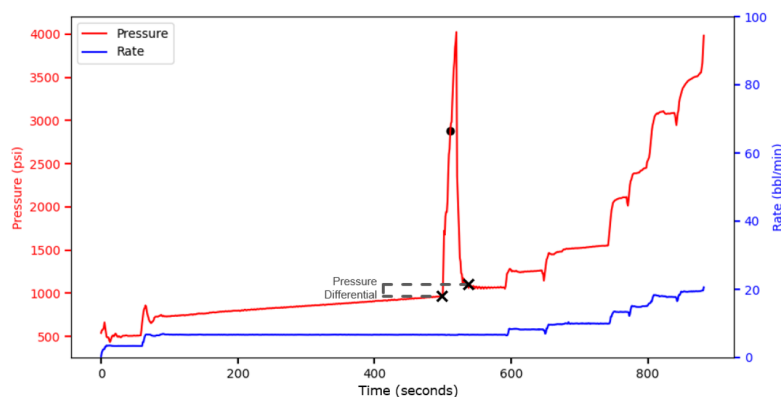


Figure 6—Pressure spike peak detection during ball landing and seating analyzed during constant rate regions.

During Treatment. During the treatment, there are three different metrics that the event detection services look for from the time treatment reaches design rate until shutdown. These are: (a) pressure comparison during pad, (b) drastic pressure drops, and (c) derivative of the drop. The algorithms identify regions where the pressure is not influenced by external factors such as rate changes, proppant concentration changes, etc. Pressure comparison during pad gets the average pressure from the time design rate is achieved until the time when proppant start at surface. Event detection in this case is straightforward, the algorithm detects the time when rate first become constant until when proppant starts. Then, from the segment of interest statistical properties are extracted. All previous stages are used in comparison with the current stage extracted statistics. Typically, for laterally homogeneous rock and similar perforation strategy (as in most unconventional shale formations), pad pressure should not vary from stage to stage. However, if there is communication with previous stimulated stages, pressure during pad can be notably lower. The reason behind this is near wellbore pressure loss can be significant during injection and pad step but if lower pressure compared to previous stages is observed it could be an indication of pumping into already eroded perforation from previous stimulated stages (Fig. 7). This metric has to be normalized for rock stress properties if significant lateral heterogeneity exists as depicted by Asif et al. (2024).

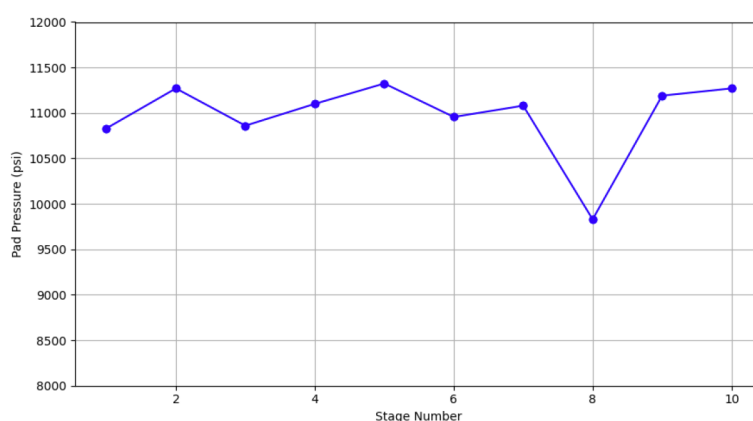


Figure 7—Average pad pressure evolution within a well showing stage 8 with a notably lower pressure compared to all the others.

The other two metrics captured during the treatment involve pressure drops at stable rate. If there are drastic pressure drops either immediate or progressive over a particular timeframe; these are detected and captured by the algorithm. These are evaluated based on magnitude of the drop. The pressure drop metric, and the drop derivative fundamental algorithm principles are similar. However, the evaluation time and magnitude of the drop vary for both metrics. The pressure drop metric looks for a sudden drastic short drop that typically implies a near-wellbore restriction, skin removal or a sudden plug slip/ plug failure. The drop derivative metric looks a sharper continuous drop over a wider period. It typically evaluates mechanical failure based on the derivative value.

The algorithm logic lies in finding constant rate regions that are not influenced by other factors. Certain operationally-dictated changes can generate drastic pressure changes that can confuse the algorithm logic. For this reason, the algorithm finds flat rate regions where the logic is similar to the rate regions from the ball seat detection module (Fig. 8). However, due to the large fluctuations caused by pumps at higher rates, data is smoothed using a moving window to minimize sharp variations point to point. In the next step, the derivatives of rate over time are clustered in groups based on a larger defined threshold.

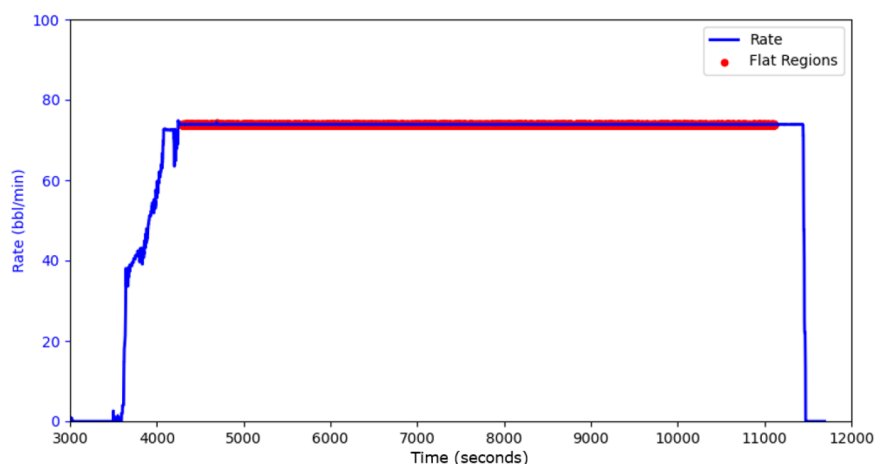


Figure 8—Flat rate region given a slurry rate during treatment where each overlapping red point represents a flat region classified instance.

After the algorithm has determined flat regions during the treatment, it focuses on the pressure behavior to capture any variations. In addition, to the fluctuations caused by sensors, pressure data is slightly smoothed using a moving window to minimize point to point variations and capture the continuous behavior. However, because of dynamic wellbore hydraulics, surface pressure behavior is affected. These include changes in chemicals concentration affecting friction pressure and changes in proppant concentration affecting hydrostatic pressure.

To capture the required pressure drop type, we search for significant change in pressure over short pressure lapses. Changes in chemicals concentration and proppant concentration do not generate such sharp sudden decreases in pressure. Here we still utilize bottomhole pressure (BHP) calculation using the novel methodology (Ugarte et al. 2025b), since it reflects surface treating pressure. Using BHP derivatives, the algorithm can understand the variations and when the variations become large above a defined threshold, the algorithm marks these regions. Each marked region is analyzed over a maximum one-minute window to ensure that the drop is significant using high and low difference bands combined with a minimum duration. This ensures removal of false positives from the algorithm logic. Then, the first and last derivatives above the defined threshold are used to find the start and end of the ‘region of interest’ to capture the delta drop as seen in Fig 9, left.

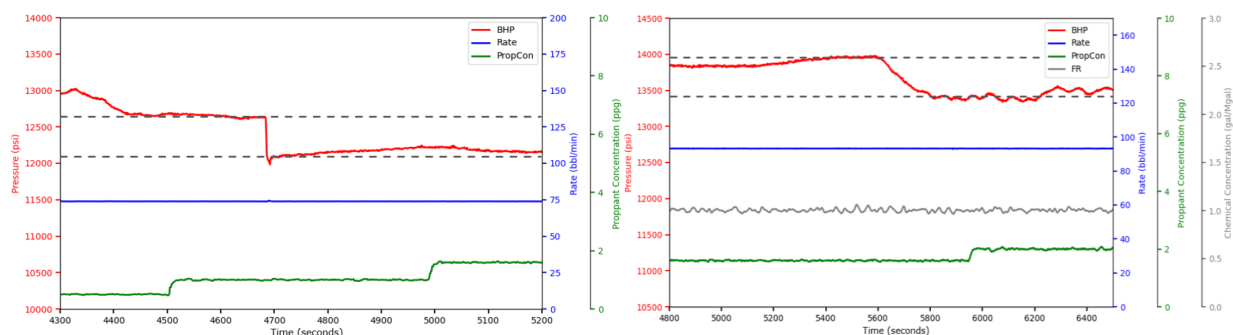


Figure 9—Mid-stage pressure drop detected from bottomhole pressure: (left) drastic pressure drop of 521 psi occurring almost instantaneously; (right) progressive pressure drop of 588 psi with a derivative of 153.4 psi/min over 3.83 minutes.

Progressive pressure drops are ones where the pressure decreases over a particular timeframe longer than the sudden pressure drop metric and utilize the derivative over the time of interest. These unexplained drops can occur during the stage and can be associated with mechanical failures such as a leak through plugs or packers. However, similar to the sudden pressure drop, for this metric we have to consider the

changes in chemicals concentration and proppant concentration confusing the algorithm logic. Hence, it is important to utilize BHP to account for the variation in hydrostatic pressure. This way the effect of changes in proppant does not generate false positives. To handle that, a change detection algorithm is used to find changes from the chemical channels of the acquired/ streaming data. Then, from the detected point using the well information we can isolate the transitional region where the new fluid or concentration is progressing through the casing volume until reaching perforations (Ugarte et al. 2025a). That marked region is ignored by the derivative evaluation and estimation. Here, we follow a similar logic as the sudden pressure drop using a smaller derivative threshold within larger valid regions (no change in rate and chemical concentration) using the bottomhole pressure as shown in Fig 9, right.

After Shutdown. At the end of the treatment, after pumps have shut down, there are three metrics related to pressure falloff analysis: Instantaneous Shut-In Pressure (ISIP); pressure decline rate; and water hammer oscillation frequency. ISIP and water hammer overlap compares the current stage with the previous stage. An overlap between these two ISIP indicate the fluid is entering the same zone. To evaluate ISIP overlap, the algorithm detects the point where rate reaches zero as the initial index time. The algorithm then takes delta time from the rate zero index to a maximum shared index time between both ISIPs. The algorithm has references of the previous stage if available to automatically compare both curves. Mean Absolute percentage error (MAPE) was used to quantify the difference between both curves. MAPE provides results in relative terms, so it is not affected by the dimension scale, rather gives a result as a percentage. If the percentage difference is below a certain threshold, it can be concluded that both stages were pumped into the same region or formation. Fig. 10 shows an example of an overlap ISIP comparison where isolation was successful (left) with the MAPE in ISIP differential was 98.2% (97.5% in the water hammer section and 99.8% in the leakoff section), and another where isolation was unsuccessful (right) where the MAPE was 87.3% (87.7% in the water hammer section and 86.8% in the leakoff section). For the decline, when available, the evaluated metric utilizes a 5-minute decline rate to compare the overlapping with previous stage. Similar to the ISIP overlap comparison but utilizes a calculated decline rate variable instead of utilizing pressure with different percentage thresholds.

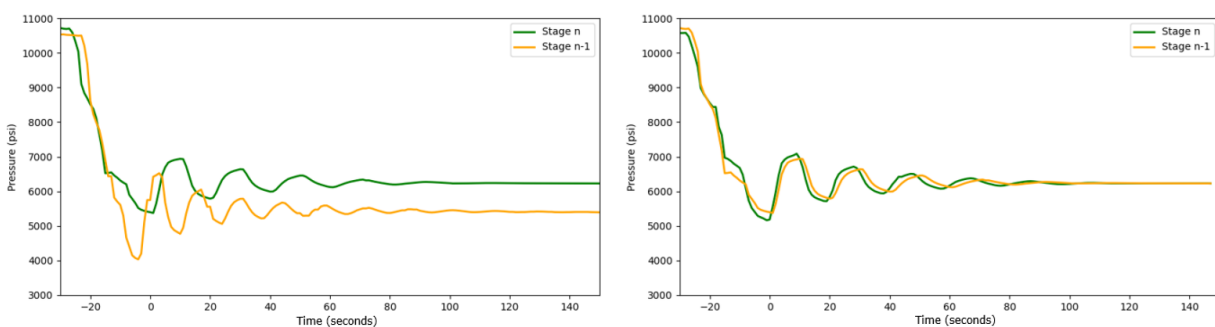


Figure 10—ISIP and initial decline overlaid curves: (left) showing no overlap; (right) overlap detected. The plots are generated in time and in specialized G-time functions based on the selection.

Finally, the last metric is the water hammer frequency. The water hammer occurs during the ISIP due to the abrupt deceleration caused by the pumps shutdown generating a shock wave oscillation from the perforation through the pipe. Similar to all the other after-shutdown metrics, water hammer frequency is a comparison with pre-treatment, injection or SRT or PDP. As established in Table 2, oscillations below four are a good signal of proper isolation while a higher number of oscillations above seven denote lack of isolation. To detect the high oscillation region, the algorithm utilizes a similar approach as Ugarte et al. (2025b) by using the second derivative of pressure to isolate the high oscillation region using a percentile as a self-adjusted threshold to account for intensity, frequency, and magnitude in the water hammer. The algorithm logic involves finding the peaks above the average of the isolated region. Fig. 11 shows some

samples of the detected oscillation by the algorithm for two cases. The peaks detected are compared to assess noticeable differences that could signal near wellbore connectivity.

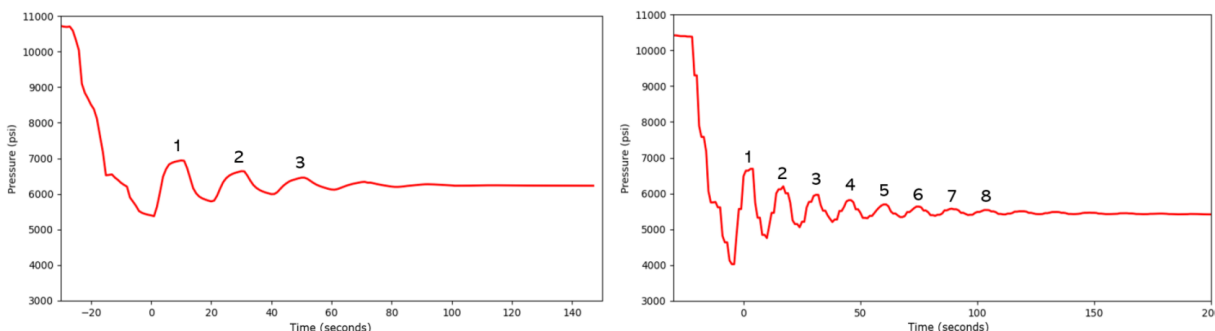


Figure 11—Water hammer frequency detection showing allocation of number of oscillations: (left) shows lower number indicating good isolation from near wellbore restrictions through new perforations; (right) high number indicating lack of isolation from extremely good near wellbore connection through eroded perforations and a conductive proppant pack.

Decision Analysis Workflow

After individual fracture pressure diagnostics have been quantified, the next essential step is to synthesize and assimilate these diverse indicators into a unified decision output (unique answer), i.e., whether the stage or interval has been hydraulically isolated or not. This is accomplished through a Multi-Criteria Decision Analysis (MCDA) framework that integrates heterogeneous diagnostic criteria, each with varying physical interpretability, signal strength, and relevance depending on operational context. [Triantaphyllou \(2020\)](#) provides a detailed summary of mathematical methods for such decision making. The approach is similar to the one taken by [Almutary et al. \(2024\)](#) where the authors calculate a placement quality index using various MCDA techniques.

The chosen MCDA methodology for this study is the Analytical Hierarchy Process (AHP) introduced by [Saaty et al. 2008](#), a structured technique for organizing and analyzing complex decisions based on mathematics and also including the subject matter expertise judgement. AHP offers the flexibility to capture subjective input (e.g., operational reliability of ball seat signal) while maintaining a rigorous comparative structure across criteria. AHP is particularly valuable in this application because it allows weighting schemes to reflect the dynamic nature of various completions architecture, treatment designs, and occurrence instance of fracture pressure indicator, rather than assuming a static global weighting.

As shown in [Fig. 12](#), the AHP implementation is segmented by completion type—Plug-and-Perf (PnP) vs. Openhole Multistage Fracturing (OH MSF), and by stimulation style—Proppant Fracturing vs. Acid Fracturing or Matrix Acidizing. This separation is essential due to variations in operational constraints and available diagnostics. For instance, PnP wells typically use ball-in-place (caged ball) isolation, making ball seat detection nearly impossible; hence, its associated criterion receives negligible weight in those scenarios. On the other hand, OH MSF designs using ball-drop mechanisms offer clear seating signatures and hence afford higher diagnostic fidelity to justify increased weighting for ball seat comparison. Similarly, if PnP wells do not use the ball-in-place, then the weighting will be increased.

The weighting distributions shown in the figures are derived from subject matter expert calibration and reflect field-prioritized experience with pressure-based diagnostics. For example, in proppant-fractured PnP wells where mid-stage pressure drops are observed, that criterion may account for more than 50% of the isolation decision logic. In contrast, in acid-stimulated OH MSF wells, decline rate and water hammer signatures may take precedence, because the reactive transport with acids governs the pressure behavior and can potentially mask the criterion output to mislead the final answer.

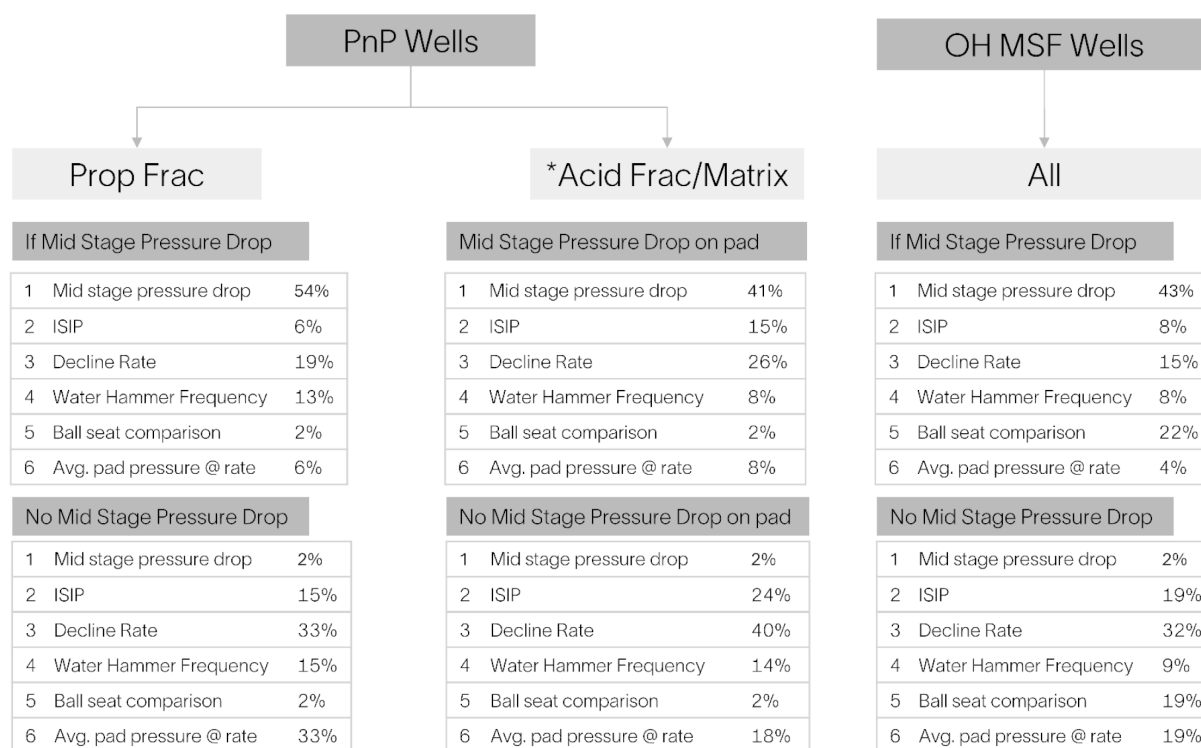


Figure 12—Criteria weighting for isolation detection based on the Analytical Hierarchy Process formulation for different scenarios. The scenarios consider treatment type and completion type to allocate correct weights.

This hierarchy-based scoring system enables a composite isolation index to be calculated for each stage, which is then mapped to a binary or confidence-scaled decision. The formulation is both interpretable and adjustable, offering engineering teams the ability to fine-tune diagnostic sensitivity.

Looking ahead, an objective extension of this approach can be realized using entropy-based weighting schemes. The Entropy Method (Zeleny, 1982) derives criteria weights directly from the magnitude and variance of pressure responses, eliminating reliance on expert judgment. Such an approach increases objectivity and enables real-time adaptive calibration of the decision framework, particularly valuable in data-rich digital operations.

In summary, the AHP-based weighting framework enables structured, field-tailored, and transparent decision-making in real-time stage isolation analysis—while allowing further evolution into data-driven adaptive schemes as operational datasets grow.

Results

The proposed fracture pressure criteria framework is able to robustly evaluate events across a stimulation stage from the beginning, through treatment, to post-shutdown diagnostics by processing the timeseries data using the different event detection algorithms designed around domain-specific operational logics. Since the intention of this paper was to focus on the novel methodology, detailed and integrated results are not presented here. A detailed case study is presented by Khan et al. (2025) validating the workflow and the MCDA algorithm integration.

During the stage beginning the ball seat detection algorithm successfully identified pressure differentials associated with ball landing events in 87% of stages with known ball seat events from a well with 15 stages (Oparin et al. 2024). The algorithm is capable of flagging differential pressure responses from pre-landing pressures to post-landing levels. The accuracy of detection was verified by visual inspection of selected events, confirming the algorithm's ability to detect valid pressure peaks, and assessing differential behavior.

The 13% cases were the stages where the ball was in place with the plug and very little volume (resulting in an insignificant pressure spike) was used to seat the ball and isolate the stage.

For example, in [figure 6](#), the algorithm successfully detected a pressure differential of 136 psi denoting successful isolation in this case. [Fig. 13](#) shows some additional ball seat detection results from the dataset. The left side figure shows a good ball seat isolation followed by a pressure drop related to rate drop. A pressure comparison (before and after ball seat) at 5-bpm yields good isolation from the algorithm logic. The ball seat detection logic becomes a reliable measurement when information regarding pumping the ball is available. The downside is information whether the ball was pumped or was in-place with the plug, is not always available and having the ball seat algorithm search the beginning of the stage without prior knowledge generates no results from a stage.

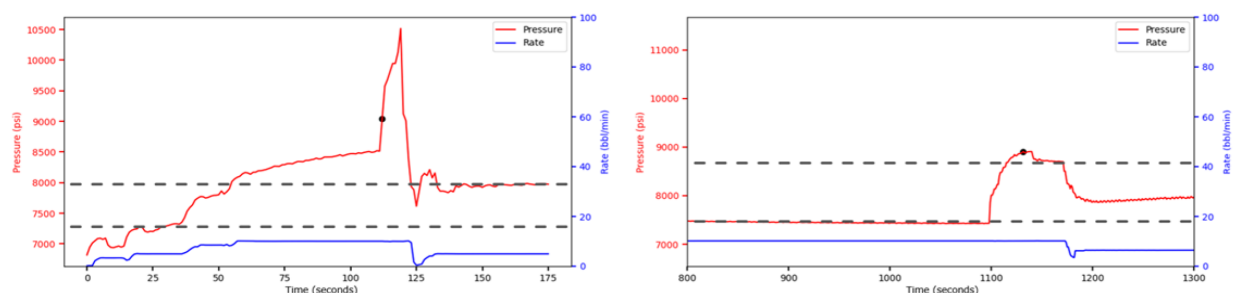


Figure 13—Ball seat results using the pressure spike peak detection during constant rate regions at the beginning of the stage: (left) $\Delta P \sim 668$ -psi at 5 bbl/min injection rate; (right) $\Delta P \sim 1,295$ -psi at 10 bbl/min injection rate.

For diagnostics during the treatment, the algorithm demonstrated high consistency and reliability in detecting pressure anomalies. The sudden and progressive pressure drops have successfully identified abrupt unexplainable losses across 100% of stages with known mid-stage pressure drop validated by the subject matter experts. Giving flexibility in detecting short sharp decreases or slightly longer progressive decreases covering a wide range of potential isolation issues such as near-wellbore restriction removal, plug slip, plug failure, plug leak, packer leak, packer failure, etc. Detected pressure drops ranged between 200 and 1000 psi. [Figures 9 and 10](#) show a sample of each case where the detected drops were 521-psi and 588-psi respectively. These events typically occurred in isolation from rate or chemical changes, indicating a potential plug slip. More results demonstrating the pressure anomalies are shown in [Fig. 14](#).

A limitation to this logic is noteworthy where the ball or plug could have potentially failed during rate increase to achieve design rate for the pad. Any pressure drop occurring during this short period is isolated from the algorithm logic due to the flat rate allocations. Hence the criticality of the overall ensemble of criteria becomes critical. The pad pressure trend comparison becomes relevant in such situations. The model availability to automatically extract average pad pressures for each stage is highly reliable. As observed in [figure 7](#), stage 8 clearly exhibits significantly lower pad pressure potentially reflecting loss of isolation when no mid stage pressure drop is found during the treatment. One of the possibilities in such a scenario is ball failure or plug slip during the rate ramp period.

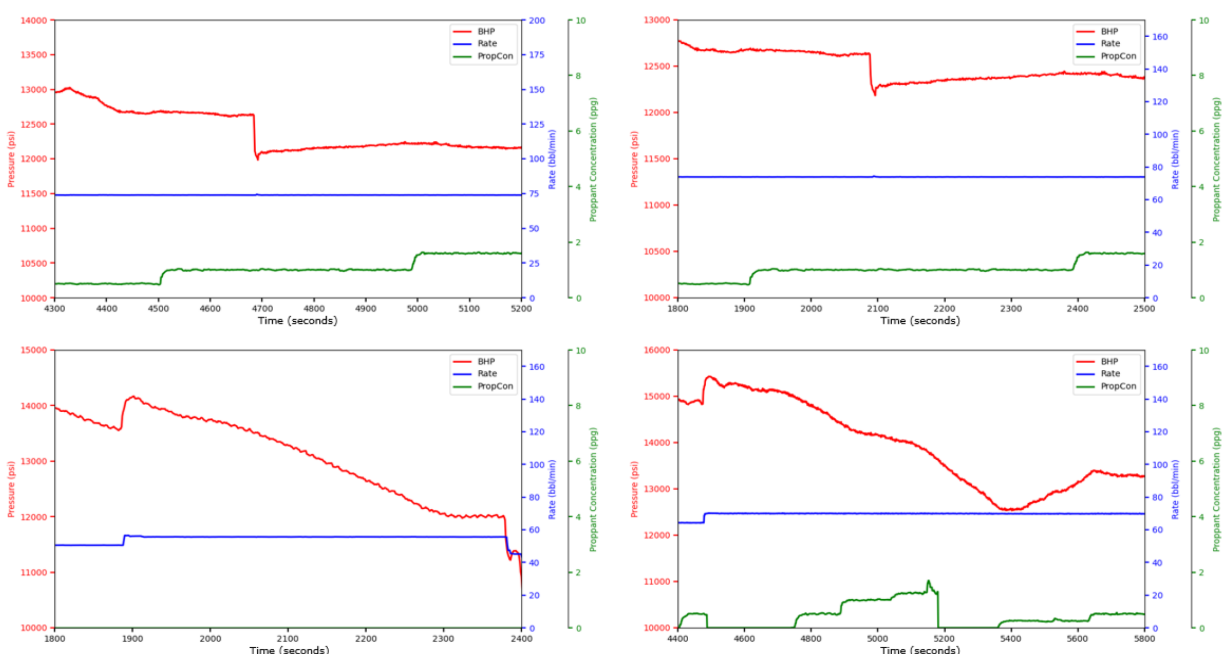


Figure 14—Pressure drops detection results observed during treatment varying from drastic to progressive pressure drops.

Post-shutdown pressure falloff analysis metrics combining the overlaps for water hammer, decline rate, and water hammer oscillations, exhibited a 93% rate of success using MAPE. Figure 11 shows two highly contrasted cases where the MAPE was 98.2% similarity for the overlapping pair and 87.3% similarity in the non-overlapping case. For these two same cases decline rate comparison using a larger window reinforced the ISIP overlap metric findings. Fig. 15 shows the ISIP comparison for each stage with its previous stage, for stages 4 through 7. It can be observed how the first two stages did not exhibit any overlap in the results, while the last two stages did show an overlap based on the results obtained using MAPE.

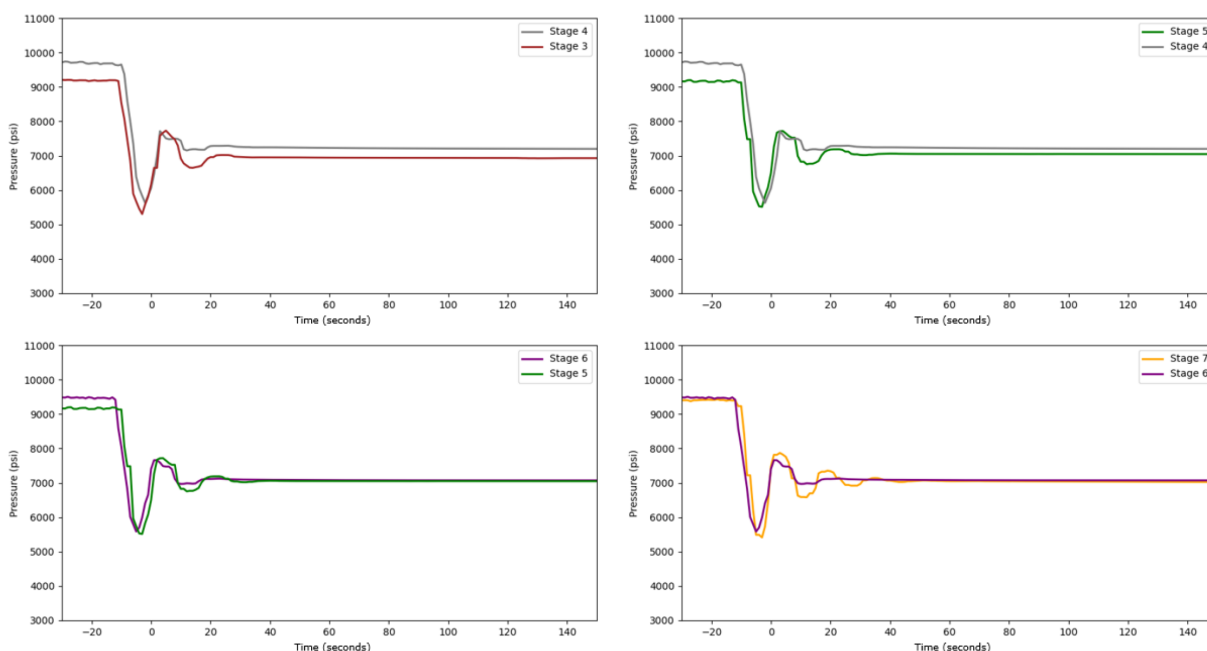


Figure 15—ISIP and decline rate overlap comparison for stages 4 through 7 compared to their respective previous. (top) results show no overlap sample in the first two stages; (bottom) overlap on the last two stages detected at the end of the treatment.

When combined, the multi-metric framework delivered a comprehensive view of isolation performance. Using the developed decision analysis, workflow provides weight to the most relevant metric signals leading to a more comprehensive results on stage isolation. Field validation against known failure cases showed agreement with the automated results, confirming the robustness and utility of the approach. The algorithms provide consistent, bias-free evaluation of stage isolation, supporting proactive diagnostics and operational decision-making in the field. Here we have not detailed the quantitative values of each metric that define isolation or lack of it. For instance, what threshold of (a) the absolute pressure drop, (b) derivative of the pressure drop, (c) MAPE for the differential in the ISIPs, (d) the MAPE in differential decline rate, etc., yields a binary independent class (isolation or no isolation) that feeds into the MCDA algorithm for the final ensemble answer? This has not been detailed here because the real-time product is tunable for these default quantitative measures by - (a) domain users in a specific geography; (b) data analytics from previous jobs where labels are provided for a good and bad isolation for a particular treatment. These aspects are detailed in our case study paper (Khan et al. 2025).

Below Table 3 provides a summary comparing the novel real-time system versus the existing plug-failure detection techniques. The current practices are based on the multiple published works that have been introduced in the earlier sections –

Table 3—Comparison of existing stage isolation diagnostics versus proposed real-time automated system.

Aspect	Conventional/existing approach	Proposed real-time automated approach
Diagnostic Criteria	Typically limited to 1–3 selected indicators depending on operator preference (e.g., ISIP overlap, water hammer, or pump-down response). Criteria usage is inconsistent across regions and service companies.	Integrates multiple criteria (ISIP overlap, decline, pad pressure, ball-seat differential, non-operational pressure drops, water hammer analytics) in an ensemble approach for holistic evaluation.
Data Integration	Manual cross-checking of separate diagnostics; operators often rely on subjective SME judgment to weigh criteria.	Advisor modularity integrated with MCDA (AHP or EM) provides standardized, logic-driven weighting tailored to completion type and stimulation method.
Real-Time Capability	Mostly post-job analysis of pressure data or delayed measurements interpretation (e.g., DAS, tracer); real-time implementation is rare.	Fully real-time, processing the acquired surface pressure and rate data with automated event detection.
Automation Level	Manual interpretation dominates; outcomes depend heavily on engineer experience and can vary between shifts or teams.	Fully automated algorithmic logic removes human subjectivity, bias and provides consistent, reproducible outputs.
Standardization	Non-standardized, operator-dependent workflows; diagnostic thresholds can vary widely within the same basin or region.	User-configurable but standardized logical thresholds allow consistent application across wells while permitting regional tuning.
Decision Turnaround	Can take few hours to 10s of hours	Outputs generated in minutes during ongoing treatment, enabling proactive adjustments.
Validation	Limited to visual SME confirmation; rarely cross-validated with multiple sources.	Validated with few edge cases in the evaluation phase through SME interpretations, then the outputs can be relied upon.
Operational Impact	Typically reactive; adjustments possible only in later stages or post-job.	Proactive decision-making possible, reducing NPT and improving stimulation efficiency in real-time.

Discussion and Way Forward

It is important to note that in this paper, the authors have focused on loss of isolation, and not just isolation plug failure, ball failure, etc. Anytime interstage communication occurs, the analysis will confirm that regardless of the type of failure which causes hydraulic communication. The mechanisms could be through plug, ball, packer, reservoir cross flow, etc. For instance, the assumption here for horizontal wells is that they are drilled in the azimuth of S_{hmin} and hence the wells have transverse fractures (which is true for most unconventional cases in North America, China, Middle East). But when wells are drilled for longitudinal

fractures, in theory, you can get communication when the fracture grows to the next zone (past the packer). This is also a form of loss of isolation through inter-stage communication.

In laterally homogenous formations where the rock quality is similar, it yields similar post-treatment ISIPs even with good isolation and that is a potential limitation of the workflow. To overcome these limitations associated with isolated anomalies (as in previous studies), multiple criteria have been added to the evaluation package and the weighting behavior is also critical. The weighting scheme can incorporate expert users' prior knowledge of the basin and treatment behavior. Similarly, the declining pressure transient can be indicative of similar rock-fluid interaction, rather than loss of isolation.

Eventually, in digital architecture, the model will be coupled with the outputs from a High Frequency Pressure Monitoring system that utilizes Tubewave velocity approach to calculate the fluid entry point as shown in [Alobaid et al. \(2022\)](#). Also, water hammer analysis in various forms has been utilized for fracture diagnostics ([Carey et al. 2015](#); [Roussel et al. 2021](#)). Within the digital environment, incorporation and coupling of various elements and techniques becomes seamless. The framework currently operates stage-by-stage, but future iterations could incorporate time-series pattern recognition across entire wells or pads using recurrent models such as LSTM networks. Additionally, integration with DAS, high-frequency strain, or real-time distributed pressure sensing could enhance detection confidence and facilitate cross-modal validation. ML-based cluster efficiency tools ([Korkin and Khan, 2024](#)) can also be used in conjunction with the proposed approach to yield a plug loss interpretation when cluster efficiency shows unreasonably high values.

Ultimately, field deployment should be followed by continuous learning, using misclassified outcomes to retrain and refine detection thresholds. This will allow the tool to evolve into a robust, adaptive analytics engine, capable of supporting diverse well architectures and stimulation workflows.

Summary and Conclusions

This study presents a novel, automated diagnostic workflow for detecting stage isolation failures and plug integrity issues in multistage fracturing. The method integrates various independent pressure analysis techniques into a cohesive logic-driven system. The criteria include ISIP correlation, pressure falloff, water hammer analysis, ball seat comparison, pressure behavioral trends, and anomaly detection. Key findings and contributions include:

- Successful real-time identification of stage communication with over 90% accuracy across field trials.
- A flexible multi-criteria decision-making (MCDM) framework based on Analytical Hierarchy Process (AHP) to tailor criteria weights by completion type and treatment fluid.
- Automated segmentation of treatment cycles and logical rule implementation without requiring post-job interpretation.
- Compatibility with existing surface pressure data, enabling lightweight and cost-effective implementation.
- Validation through SME feedback loop, confirming both reliability and operational relevance.

The proposed system has the potential to significantly reduce NPT, improve stimulation efficiency, and support proactive decision-making during stimulation. By bridging the gap between high-resolution diagnostics and real-time field implementation, this work marks a shift toward more intelligent and automated well integrity monitoring during hydraulic fracturing operations.

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